Sightline `Case demo-full`

Reconstruction analysis report

Car into turning tractor-semitrailer at an uncontrolled median crossover, US-27A near Williston, Florida (2016) — full case file

demo-full

Case reference	demo-full
Overall result	world patched · 5 findings (1 contradiction, 4 consistent)
Reconstruction	world_v4 (version 4), 2 review rounds
Validation	8 of 12 published facts reproduced within tolerance (NTSB HAR-17/02)
Run mode	live — model outputs generated live
Report generated	October 25, 2026 at 4:04 PM PDT

Analysis draft for expert review — not evidence

This document is a working analysis prepared to help a qualified accident-reconstruction expert review the testimony in this case. It is not an expert opinion and is not offered as evidence. Every value in the reconstruction is tied to a citation; values marked *inferred* or *assumed* are modeling choices that an expert must confirm.

Kinematic reconstruction: motion fitted to cited evidence; not a crash-physics simulation.

Contents

1 Summary of findings
2 Evidence received
3 Findings
4 Exhibits (figures)
5 Independent checks
6 Validation against NTSB HAR-17/02
7 Methodology
8 Provenance

1 Summary of findings

No.	Status	Finding	Severity
1	contradiction	Witness B's pen mark #2 on Exhibit 4 places her at [8.5, 0, 9.0] m (immediately against the south back of the bench shelter), where both the bench shelter and the turning tractor-semitrailer block her claimed continuous view of Vehicle A right up until impact.	high
2	consistent	Driving witnesses A and C were modeled with static eye coordinates rather than riding with their vehicles (`vehicleId` and `driverEye` pose updates in `setTime`), and Witness A's observationTime was set to t=7.0s (when the tractor already blocked his view) instead of during his clear-view window before the trailer entered the eastbound lanes.	medium
3	consistent	Witness D's testimony that she was inside the retail store behind the register and could not see the highway or the collision is physically consistent with the reconstruction.	low
4	consistent	Roadway geometry, vertical crest profile, vehicle speeds, impact alignment, and post-impact trajectory match the NTSB factual reports and crash reconstruction.	low
5	consistent	Scene lighting, clear weather, dry pavement, sun position (azimuth 265.0°, elevation 45.1°), and absence of glare or active headlights match the documentary record and solar audit; no real scene footage exists.	low

2 Evidence received

Sizes and running times are read from the files as stored. Hashes are SHA-256 of the file given to the model (first 16 hex digits; full values in Section 8).

Exh.	File	Type	Size	Length	SHA-256
E-1	deposition_witness_B.mp4	Video	19.0 MB 19,910,361 bytes	2:00:00 7,200.0 s	5c9dc19f396dd976
E-2	deposition_witness_D.mp4	Video	7.0 MB 7,313,799 bytes	45:00 2,700.0 s	9f26ce5d5e88ad0a
E-3	exhibit4_intersection_map.png	Image	53.3 KB 54,576 bytes	—	e44f99efb5dd4830

3 Findings

Finding 1 contradiction

Severity: high · ref F1

Witness B's pen mark #2 on Exhibit 4 places her at [8.5, 0, 9.0] m (immediately against the south back of the bench shelter), where both the bench shelter and the turning tractor-semitrailer block her claimed continuous view of Vehicle A right up until impact.

In her deposition, Witness B testified that she stood outside in front of the store and had 'a clear view of the car the whole way in. I watched it right up until the moment it hit. Nothing was in the way' (Depo Witness B 41:10-41:20, 52:00-52:07, 56:00-56:18). However, when asked to mark her exact standing position on Exhibit 4 (Depo Witness B 1:54:24-1:54:50; frame at 1:54:34), the center of her red circled-X pen mark (#2) falls directly on the south edge of the bench shelter rectangle (which spans X = 6.75..10.25 m, Z = 7.45..8.95 m, centered at [8.5, 8.2] m per Exhibit 4's 40 m scale bar and AOI origin [0, 0]). Measured with Exhibit 4's scale bar (8.025 px/m), her mark #2 is at [8.5, 0, 9.0] m—not [7.5, 0, 12.0] m as coded in Round 3. Furthermore, her claim 'watched it right up until the moment it hit' requires evaluating her sightline in the last second before the t=10.43s impact (t=9.9s, as shown in overhead_t9p9.png), not at t=9.5s before the truck's cab and trailer interpose. From [8.5, 0, 9.0] m, the sheet-metal back of the bench shelter (Depo Witness D 10:16-10:29) and the turning tractor-semitrailer (veh_truck_tractor and veh_semitrailer) cross directly between Witness B and Vehicle A during the final second before impact (t ≈ 9.6s to 10.43s), making her claim physically impossible. The world module is patched to place Witness B at [8.5, 0, 9.0] m with observationTime = 9.9 s.

Evidence cited

1. evidence/deposition_witness_B.mp4 — Depo Witness B 41:10-41:20, 52:00-52:07, 56:00-56:18, 1:54:24-1:54:50 (frame 1:54:34)
2. evidence/exhibit4_intersection_map.png — Scale bar 0-40 m and AOI reference at [0,0]; bench shelter centered at [8.5, 8.2] m; pen mark #2 centered at [8.5, 9.0] m
3. evidence/deposition_witness_D.mp4 — Depo Witness D 09:43-10:29 (bench shelter has roof and sheet-metal back)
4. Render — overhead_t9p9.png @ t=9.9s; wit_witness_B_t9p5.png @ t=9.5s
5. Independent check — sightline wit_witness_B→veh_car_a @ t=9.5s: clear at [7.5, 1.57, 12.0] before tractor-trailer and shelter interpose at [8.5, 1.57, 9.0] @ t=9.9s

Independent check results

Check	Result	Finding of the check
Line of sight Witness B (Store Pedestrian)	blocked	Line of sight Witness B (Store Pedestrian) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 9.90 s: **Blocked by Bench shelter in front of retail store (roof on posts with sheet-metal back panel)** (target 29.9 m away). Robustness: blocked from **25 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — robust (Bench shelter in front of retail store (roof on posts with sheet-metal back panel) 11, Vehicle T (Tractor): 2014 conventional truck-tractor turning left from WB US-27A onto SB NE 140th Ct 14). Gaze: target **29.5°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.

See Figures 3A, 3B.

Finding 2 consistent

Severity: medium · ref F2

Driving witnesses A and C were modeled with static eye coordinates rather than riding with their vehicles (`vehicleId` and `driverEye` pose updates in `setTime`), and Witness A's observationTime was set to t=7.0s (when the tractor already blocked his view) instead of during his clear-view window before the trailer entered the eastbound lanes.

Witness A was driving a black pickup truck westbound at ~60 mph (~0.25 mi behind the tractor-trailer) and testified that he saw Vehicle A crest the hill and descend for 2–3 seconds before the turning semitrailer entered the eastbound lanes and blocked his view (docket_har1702.txt p.11-12, p.29-30; docket_hp_att4_witness_interview.txt p.5-6, p.12, p.24-25; witness_A.md p.1). In the Round 3 world module, Witness A's eye position was fixed at [257.26, 0, -34.75] without `vehicleId: 'veh_witness_a_pickup'` or dynamic updates in `setTime(t)`, and his `observationTime` was set to t=7.0s, where the independent check showed `veh_truck_tractor` already blocking the ray to `veh_car_a` as the truck crossed the median. Similarly, Witness C (stopped in his pickup at [-18.0, 0, -31.09] m) lacked `vehicleId: 'veh_witness_c_pickup'` and dynamic `driverEye` updates in `setTime(t)`. In the patched world, both witnesses are bound to their vehicles via `vehicleId` and updated in `setTime(t)` from `vehicle.userData.driverEye`, and Witness A's `observationTime` is set to t=4.0s (during the 2–3 s window after Vehicle A crests the hill and before Vehicle T's turn blocks his sightline, consistent with docket_vehicle_dynamics_study.txt p.8 Table 2).

Evidence cited

1. text/docket_har1702.txt — p.11-12, p.29-30
2. text/docket_hp_att4_witness_interview.txt — p.5-6, p.12, p.16, p.24-25
3. text/docket_vehicle_dynamics_study.txt — p.2, p.4, p.8 Table 2 (Witness Line of Sight Time 3.2-6.8 s before truck obscures view)
4. evidence/witness_A.md — p.1
5. evidence/witness_C.md — p.1
6. Independent check — sightline wit_witness_A→veh_car_a @ t=7s: blocked by veh_truck_tractor; sightline wit_witness_C→veh_semitrailer @ t=10.43s: pass

Independent check results

Check	Result	Finding of the check
Line of sight Witness A (Westbound Driver)	clear	Line of sight Witness A (Westbound Driver) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 4.00 s: **Clear** (target 553.6 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **2.3°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.

Check	Result	Finding of the check
Line of sight Witness C (Stopped Westbound Driver)	clear	Line of sight Witness C (Stopped Westbound Driver) → Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front) at t = 10.43 s: **Clear** (target 35.0 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **6.6°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.

See Figures 2A, 2B, 4A, 4B.

Finding 3 consistent

Severity: low · ref F3

Witness D's testimony that she was inside the retail store behind the register and could not see the highway or the collision is physically consistent with the reconstruction.

Witness D testified that at the time of the crash she was inside the store on the southeast corner of US-27A and NE 140th Court, standing behind the register restocking cigarettes, and that the drink cooler, shelving, and window posters blocked her view of the highway so that she only heard the loud bang and came outside 10–15 seconds later (Depo Witness D 06:39-08:47, 13:39-14:45, 18:02-19:22, 41:40-42:18). The independent sightline check from `wit_witness_D` at [24.0, 1.6, 15.0] m to `veh_car_a` at t=10.43s confirms that `bld_retail_store` blocks her line of sight.

Evidence cited

1. evidence/deposition_witness_D.mp4 — Depo Witness D 06:39-08:47, 13:39-14:45, 18:02-19:22, 41:40-42:18
2. Independent check — sightline wit_witness_D→veh_car_a @ t=10.43s: blocked by bld_retail_store
3. Render — overhead_t9p9.png @ t=9.9s

Independent check results

Check	Result	Finding of the check
Line of sight Witness D (Store Counter Clerk)	partial	Line of sight Witness D (Store Counter Clerk) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 10.43 s: **Blocked by Retail store / country store on SE corner of US-27A & NE 140th Court (solid west wall, posters/cooler on north windows)** (target 30.4 m away). Robustness: blocked from **23 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — partial (Retail store / country store on SE corner of US-27A & NE 140th Court (solid west wall, posters/cooler on north windows) 23). Gaze: target **2.9°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.

See Figures 5A, 5B.

Finding 4 consistent

Severity: low · ref F4

Roadway geometry, vertical crest profile, vehicle speeds, impact alignment, and post-impact trajectory match the NTSB factual reports and crash reconstruction.

US-27A is modeled with four 12-ft (3.66 m) lanes, a 78-ft (23.77 m) median/turn-lane separation, a 130-ft (39.6 m) crossover, and a vertical crest 1,132 ft (345.0 m) west of the intersection descending at 2.15% (docket_har1702.txt p.13, p.21-22, p.29; docket_tech_recon_factual.txt p.3-4). Vehicle A approaches eastbound at a constant 74 mph (33.08 m/s) with no pre-impact braking skids, striking the right side of the 53-ft semitrailer 45 ft 5 in (13.84 m) behind the tractor front at t=10.43s, then departing the right roadside at ~1.7° (326 ft / 99.4 m), breaching two wire fences, severing the utility pole at 855 ft (260.6 m) at t=18.83s (8.4 s after impact), and coming to rest at 910 ft (277.4 m) (docket_har1702.txt p.11, p.15-16; docket_tech_recon_factual.txt p.4-5, p.12-13). All independent plausibility and timing checks pass.

Evidence cited

1. text/docket_har1702.txt — p.11, p.13, p.15-16, p.21-22, p.29
2. text/docket_tech_recon_factual.txt — p.3-5, p.10-13
3. Independent check — plausibility veh_car_a, veh_truck_tractor, veh_semitrailer, veh_witness_a_pickup: pass; timing veh_car_a→veh_semitrailer: availableS 10.43s (sufficient)

Independent check results

Check	Result	Finding of the check
Perception-reaction time Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off)	sufficient	Perception-reaction time: Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) had **10.43 s** from first sight of Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front) to impact (first visible t = 0.00 s); PRT range 1.0–2.5 s, typical 1.5 s — **sufficient**. Time from the conflict first arising to impact: 1.73 s.
Plausibility Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off)	pass	Kinematic plausibility: peak 0.03 g at t = 1.70 s, peak speed 74.0 mph. No issues.
Plausibility Vehicle T (Tractor): 2014 conventional truck-tractor turning left from WB US-27A onto SB NE 140th Ct	pass	Kinematic plausibility: peak 0.99 g at t = 2.65 s, peak speed 15.4 mph. No issues.

Check	Result	Finding of the check
Plausibility Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front)	pass	Kinematic plausibility: peak 0.87 g at t = 4.85 s, peak speed 14.1 mph. No issues.

Finding 5 consistent

Severity: low · ref F5

Scene lighting, clear weather, dry pavement, sun position (azimuth 265.0°, elevation 45.1°), and absence of glare or active headlights match the documentary record and solar audit; no real scene footage exists.

The crash occurred on May 7, 2016 at 16:36:12.7 EDT (UTC-04:00) at 29.4106° N, -82.53979° W in clear daylight with dry roads (docket_har1702.txt p.11, p.15, p.32 fn.40; docket_tech_recon_factual.txt p.16; police_report.md p.1). The NOAA solar calculation and the NTSB Technical Reconstruction Group Factual Report (p.16: sun azimuth 264.2°–265.6°, elevation 46.4°–44.2°) match the rendered sun at azimuth 265.0° and elevation 45.1° (0.0° error; shadow bearing 85° E). Vehicle A's headlights are off in accordance with the downloaded ECU data (docket_driver_assistance_factual.txt p.7-8; docket_har1702.txt p.15). No witness or driver has `glareRisk: true`; the smallest sun-to-sight-line angle for any witness is 45.2° (`wit_witness_A`), and the smallest sun-to-travel-direction angle for any driver is 45.1° (`veh_witness_a_pickup`), consistent with Witness A's and Witness B's testimony that the sun was high and not in their eyes (docket_har1702.txt p.12, p.30; Depo Witness B 23:44-24:01, 34:19-34:27). No real scene footage exists to compare lighting against; lighting was compared against the documentary record and computed sun position only.

Evidence cited

1. Lighting audit — expected sun az 265.0 / el 45.1 vs render az 265.0 / el 45.1 (error 0.0 deg); shadow bearing 85 deg; min witness sun angle 45.2 deg (glareRisk false); realFootage false
2. text/docket_tech_recon_factual.txt — p.16 (Astronomical data: clear skies, sun az 264.2-265.6 deg, el 46.4-44.2 deg)
3. text/docket_har1702.txt — p.12, p.15 (daylight, clear, dry, headlights off, sun glare not a problem)
4. Render — wit_witness_B_t9p5.png @ t=9.5s; overhead_t9p9.png @ t=9.9s

Independent check results

Check	Result	Finding of the check
Line of sight Witness A (Westbound Driver)	clear	Line of sight Witness A (Westbound Driver) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 4.00 s: **Clear** (target 553.6 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **2.3°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.
Line of sight Witness B (Store Pedestrian)	blocked	Line of sight Witness B (Store Pedestrian) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 9.90 s: **Blocked by Bench shelter in front of retail store (roof on posts with sheet-metal back panel)** (target 29.9 m away). Robustness: blocked from **25 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — robust (Bench shelter in front of retail store (roof on posts with sheet-metal back panel) 11, Vehicle T (Tractor): 2014 conventional truck-tractor turning left from WB US-27A onto SB NE 140th Ct 14). Gaze: target **29.5°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.
Line of sight Witness C (Stopped Westbound Driver)	clear	Line of sight Witness C (Stopped Westbound Driver) → Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front) at t = 10.43 s: **Clear** (target 35.0 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **6.6°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.
Line of sight Witness D (Store Counter Clerk)	partial	Line of sight Witness D (Store Counter Clerk) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 10.43 s: **Blocked by Retail store / country store on SE corner of US-27A & NE 140th Court (solid west wall, posters/cooler on north windows)** (target 30.4 m away). Robustness: blocked from **23 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — partial (Retail store / country store on SE corner of US-27A & NE 140th Court (solid west wall, posters/cooler on north windows) 23). Gaze: target **2.9°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°.

See Figures 2A, 2B, 3A, 3B, 4A, 4B, 5A, 5B.

4 Exhibits (figures)

Figures are rendered from the 3D reconstruction at the time each observation is tested. Eye-level views use a 60° camera at the witness's marked position and eye height; the banner states the ray-cast result. Renders illustrate geometry only; they do not show what a person perceived.



Figure 1 Overhead plan (north up) at t = 9.90 s with every witness's sight line to the object they describe; green = clear, red = blocked.

Source render: figs/overview.png · reconstruction world_v4

Witness A (Westbound Driver) · observation at t = 4.00 s · docket_har1702.txt p.11-12, p.29-30; **docket_hp_att4_witness_interview.txt** p.5-6, p.12; **docket_vehicle_dynamics_study.txt** p.2, p.4, p.8 **Table 2; witness_A.md** p.1



Figure 2A Overhead plan (north up) at t = 4.00 s showing only Witness A (Westbound Driver)'s sight line to Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off); green = clear, red = blocked, dashed grey = occluded remainder.

Source render: figs/wit_witness_A_overhead.png · reconstruction world_v4



Figure 2B Witness A (Westbound Driver) — eye-level view from the marked position (eye height 1.6 m) at t = 4.00 s, the time of the observation tested, looking toward Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off): line of sight clear; blocked from 0 of 25 positions within 2 m.

Source render: figs/wit_witness_A_eye.png · reconstruction world_v4

Witness B (Store Pedestrian) · observation at t = 9.90 s · Depo Witness B 17:01 (5 ft 6 in tall), 41:10-41:20, 52:00-52:07 ("watched it the whole way, right up until it hit"), 1:54:34 (Exhibit 4 pen mark #2 centered at [8.5, 0, 9.0] m on south edge of bench shelter)

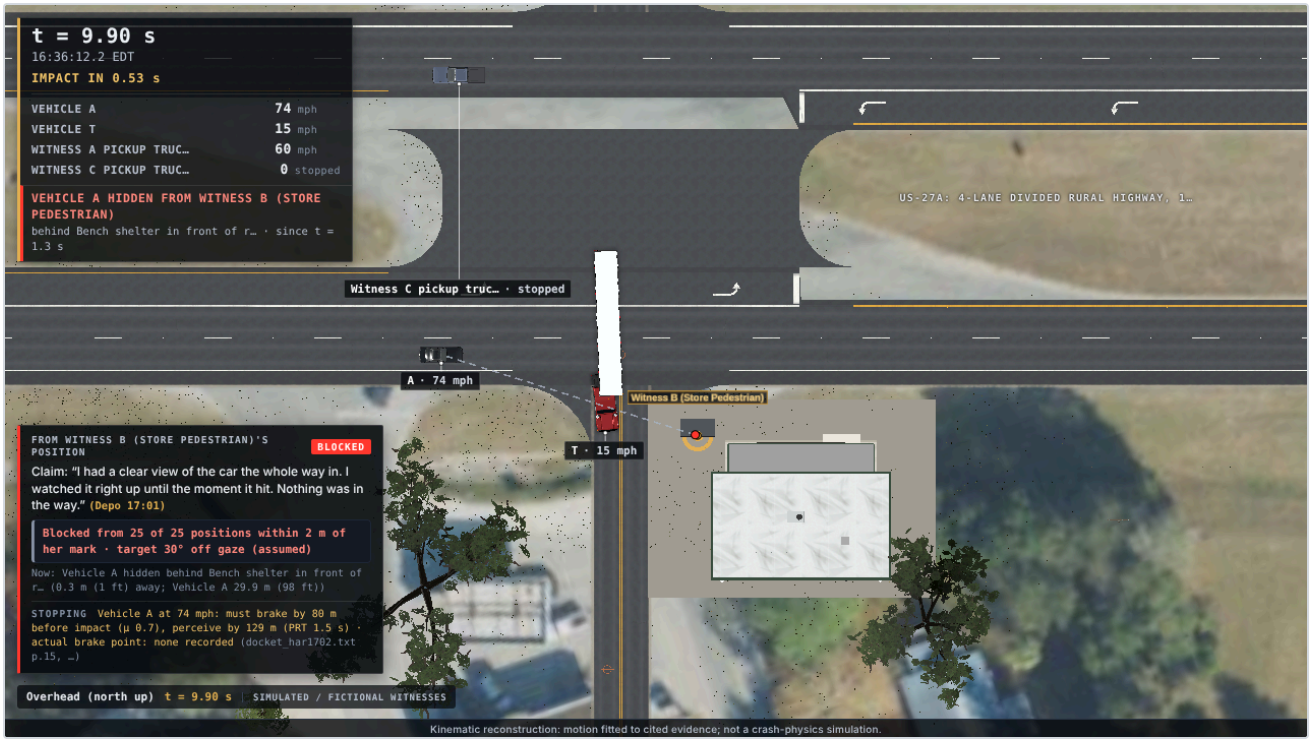


Figure 3A Overhead plan (north up) at t = 9.90 s showing only Witness B (Store Pedestrian)'s sight line to Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off); green = clear, red = blocked, dashed grey = occluded remainder.

Source render: figs/wit_witness_B_overhead.png · reconstruction world_v4



Figure 3B Witness B (Store Pedestrian) — eye-level view from the marked position (eye height 1.6 m) at t = 9.90 s, the time of the observation tested, looking toward Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off): line of sight blocked by Bench shelter in front of retail store (roof on posts with sheet-metal back panel); blocked from 25 of 25 positions within 2 m.

Source render: figs/wit_witness_B_eye.png · reconstruction world_v4

Witness C (Stopped Westbound Driver) · observation at t = 10.43 s · witness_C.md p.1**; police_report.md** p.1



Figure 4A Overhead plan (north up) at t = 10.43 s showing only Witness C (Stopped Westbound Driver)'s sight line to Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front); green = clear, red = blocked, dashed grey = occluded remainder.

Source render: figs/wit_witness_C_overhead.png · reconstruction world_v4



Figure 4B Witness C (Stopped Westbound Driver) — eye-level view from the marked position (eye height 1.6 m) at t = 10.43 s, the time of the observation tested, looking toward Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front): line of sight clear; blocked from 0 of 25 positions within 2 m.

Source render: figs/wit_witness_C_eye.png · reconstruction world_v4

Witness D (Store Counter Clerk) · observation at t = 10.43 s · Depo Witness D 07:38-08:47, 13:39-14:45, 18:02-19:22, 41:40-42:18



Figure 5A Overhead plan (north up) at t = 10.43 s showing only Witness D (Store Counter Clerk)'s sight line to Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off); green = clear, red = blocked, dashed grey = occluded remainder.

Source render: figs/wit_witness_D_overhead.png · reconstruction world_v4



Figure 5B Witness D (Store Counter Clerk) — eye-level view from the marked position (eye height 1.6 m) at t = 10.43 s, the time of the observation tested, looking toward Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off): line of sight blocked by Retail store / country store on SE corner of US-27A & NE 140th Court (solid west wall, posters/cooler on north windows); blocked from 23 of 25 positions within 2 m.

Source render: figs/wit_witness_D_eye.png · reconstruction world_v4

5 Independent checks

These checks are deterministic geometry and physics computed from the reconstruction; they are not model output. Line of sight is a ray cast from the witness's eye to the target, repeated from 25 deterministic positions in a 2 m disc around the marked position (eye height ±0.1 m) to test sensitivity to where the witness stood. Perception-reaction time compares the time available against a 1.0–2.5 s range. Plausibility flags accelerations over 1.0 g outside ±0.5 s of impact, jumps over 5 m per 0.05 s and speeds over 60 m/s.

Check	Result	Finding of the check
Line of sight Witness A (Westbound Driver)	clear	Line of sight Witness A (Westbound Driver) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 4.00 s: **Clear** (target 553.6 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **2.3°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°. Basis: docket_har1702.txt p.11-12, p.29-30; docket_hp_att4_witness_interview.txt p.5-6, p.12; docket_vehicle_dynamics_study.txt p.2, p.4, p.8 Table 2; witness_A.md p.1
Line of sight Witness B (Store Pedestrian)	blocked	Line of sight Witness B (Store Pedestrian) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 9.90 s: **Blocked by Bench shelter in front of retail store (roof on posts with sheet-metal back panel)** (target 29.9 m away). Robustness: blocked from **25 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — robust (Bench shelter in front of retail store (roof on posts with sheet-metal back panel) 11, Vehicle T (Tractor): 2014 conventional truck-tractor turning left from WB US-27A onto SB NE 140th Ct 14). Gaze: target **29.5°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°. Basis: Depo Witness B 17:01 (5 ft 6 in tall), 41:10-41:20, 52:00-52:07 ("watched it the whole way, right up until it hit"), 1:54:34 (Exhibit 4 pen mark #2 centered at [8.5, 0, 9.0] m on south edge of bench shelter)
Line of sight Witness C (Stopped Westbound Driver)	clear	Line of sight Witness C (Stopped Westbound Driver) → Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front) at t = 10.43 s: **Clear** (target 35.0 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **6.6°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°. Basis: witness_C.md p.1; police_report.md p.1

Check	Result	Finding of the check
Line of sight Witness D (Store Counter Clerk)	partial	Line of sight Witness D (Store Counter Clerk) → Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) at t = 10.43 s: **Blocked by Retail store / country store on SE corner of US-27A & NE 140th Court (solid west wall, posters/cooler on north windows)** (target 30.4 m away). Robustness: blocked from **23 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — partial (Retail store / country store on SE corner of US-27A & NE 140th Court (solid west wall, posters/cooler on north windows) 23). Gaze: target **2.9°** off the line of gaze (gaze assumed toward Initial Point of Impact (POI) in EB outside lane of US-27A at NE 140th Court); not peripheral (30° or less); camera FOV 60°. Basis: Depo Witness D 07:38-08:47, 13:39-14:45, 18:02-19:22, 41:40-42:18
Perception-reaction time Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off)	sufficient	Perception-reaction time: Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off) had **10.43 s** from first sight of Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front) to impact (first visible t = 0.00 s); PRT range 1.0–2.5 s, typical 1.5 s — **sufficient**. Time from the conflict first arising to impact: 1.73 s. Basis: meta.impactT
Plausibility Vehicle A: 2015 4-door passenger sedan (eastbound at 74 mph on TACC + Autosteer, headlights off)	pass	Kinematic plausibility: peak 0.03 g at t = 1.70 s, peak speed 74.0 mph. No issues. Basis: docket_har1702.txt p.11, p.15; docket_highway_att1_fl_crash_report.txt p.2 (BLK 4D); Depo Witness B 40:51
Plausibility Vehicle T (Tractor): 2014 conventional truck-tractor turning left from WB US-27A onto SB NE 140th Ct	pass	Kinematic plausibility: peak 0.99 g at t = 2.65 s, peak speed 15.4 mph. No issues. Basis: docket_har1702.txt p.11, p.16 fn.6; docket_highway_att1_fl_crash_report.txt p.2 (2014 FRHT CASCADIA RED)
Plausibility Vehicle T (Semitrailer): 2003 53-ft white refrigerated van semitrailer (struck on right side 45 ft 5 in behind tractor front)	pass	Kinematic plausibility: peak 0.87 g at t = 4.85 s, peak speed 14.1 mph. No issues. Basis: docket_har1702.txt p.11, p.16-17; docket_tech_recon_factual.txt p.10-11; Depo Witness B 1:01:43

Check	Result	Finding of the check
Plausibility Witness A pickup truck: black pickup traveling westbound on US-27A at ~60 mph (~0.25 mi behind truck)	pass	Kinematic plausibility: peak 0.61 g at t = 19.35 s, peak speed 60.6 mph. No issues. Basis: docket_har1702.txt p.11-12, p.29-30; docket_hp_att4_witness_interview.txt p.5-6, p.16; witness_A.md p.1

Skid-to-speed: not computed — the case contains no pre-impact tire marks to measure.

6 Validation against NTSB HAR-17/02

The reconstruction (world_v4.js) was measured against 12 facts stated in the published investigation. **8 of 12** are reproduced within tolerance. Facts not reproduced are reported, not hidden.

Published fact	Published	Model	Tolerance	Result	Citation
Car speed just before impact mean over t=9.35..10.3 s	74 mph	74 mph	±2 mph	match	NTSB/HAR-17/02 p.2, p.15 (facts.md F17)
No braking before impact (max deceleration in the 5 s before impact) over t=5.35..10.3 s, 0.5 s smoothing	0 g	0 g	±0.1 g	match	NTSB/HAR-17/02 p.15 (facts.md F18); p.5 no pre-collision roadway marks (F22)
Time for the car to cover the 1,132 ft from the crest to impact (NTSB mutual sight time) car path starts only 342 m (10.35 s) before impact; cannot cover 345 m	10.40 s	not reproduced	±0.3 s	no match	NTSB/HAR-17/02 p.29 (facts.md F28)
Vertical crest west of the intersection (sight restriction) - distance from impact point to the crest occluder's near edge impact point to the occluder edge nearest the intersection, along the car's travel direction	345 m	339 m	±15 m	match	NTSB/HAR-17/02 p.22-23, p.29 (facts.md F10, F27): 1,132 ft

Published fact	Published	Model	Tolerance	Result	Citation
Impact in the right (outside) eastbound lane at the NE 140th Ct crossing (car front at impact vs. case-frame POI) car front at t=10.35s	(0.00, 0.00) m	(-2.62, 0.00) m	x ±5, z ±1.83 m	match	NTSB/HAR-17/02 p.3, p.5 (facts.md F21); lane 12 ft (F6) -> +/-1.83 m lateral; origin = POI per case.json coordinate_frame_note (modeling convention)
Impact point on the trailer, distance behind the front of the tractor at t=10.35s	13.84 m	11.56 m	±1 m	no match	NTSB/HAR-17/02 p.6 fn.6 (facts.md F16): 45 ft 5 in; tolerance = half the car's width
Tractor-semitrailer combination length at t=0s (tractor/trailer 0 deg apart)	21.82 m	20.11 m	±0.5 m	no match	NTSB/HAR-17/02 p.1, p.6 (facts.md F15): 71 ft 7 in
Time of day of the collision	16:36:12.7	16:36	±60 s	match	NTSB/HAR-17/02 p.1, p.15 (facts.md F1): car data recorded collision at 4:36:12.7 p.m.
Pole strike (airbag deployment) time after first impact	8.40 s	0.08 s	±0.5 s	no match	NTSB/HAR-17/02 p.5 (facts.md F26); pole struck on post-impact path p.1 (F24)
Car travel from point of impact to final rest straight line, impact point to car centre at end	277 m	280 m	±20 m	match	NTSB/HAR-17/02 p.1, p.3 (facts.md F25; police_report p.3 'POI to final rest'): about 910 ft
Final rest off the road to the right (south) of the eastbound lanes	>= 3.35	21 m	±0 m	match	NTSB/HAR-17/02 p.1, p.3 (facts.md F23, F25: left the right roadside, rest in a front yard); 12-ft lane half-width + 5-ft right shoulder (F6) = 3.35 m from lane centre
Final rest perpendicular to the highway	90 deg	90 deg	±20 deg	match	NTSB/HAR-17/02 p.1, p.3 (facts.md F25)

7 Methodology

1. **Ingest.** Each evidence file is stored unchanged; text is extracted from documents and the deposition transcript is kept with its timecodes so that every statement can be cited as `Depo MM:SS`.
2. **Reconstruction.** A multimodal model (gemini-4-argon) reads the evidence and writes a time-parameterised 3D scene: road geometry, vehicles and their paths, occluders, and each witness's marked position, eye height and observation time. Every object and value carries a citation to the evidence it came from; values without direct support are labelled *inferred* or *assumed*.
3. **Review rounds.** The scene is rendered from overhead and from each witness's eye position. The model reviews its own renders against the evidence and revises the scene; this case went through 2 rounds. Each version is recorded with its inputs and prompt (Section 8).
4. **Independent checks.** Deterministic code, separate from the model, ray-casts every witness's sight line (with a 25-position robustness test), measures how far the target was from the witness's gaze, computes perception-reaction time available, skid-to-speed ranges where marks exist (μ 0.6–0.8), and flags physically implausible motion.
5. **Validation.** Where a published investigation exists, the reconstruction is measured against its stated facts (Section 6).
6. **Findings.** Each finding states a status (contradiction, consistent, unverifiable), cites the testimony and documents it rests on, and lists the independent checks that bear on it.

Limitations

Kinematic reconstruction: motion fitted to cited evidence; not a crash-physics simulation. Sight lines are geometric: a clear line of sight does not establish that a witness noticed an object, and gaze direction is assumed where testimony does not state it. Results depend on the marked witness position, which the robustness test brackets to 2 m. Conclusions must be confirmed by a qualified reconstructionist before any use.

8 Provenance

Version	Mode	Model / prompt	Created	Output SHA-256
world_v1 v1	live	gemini-4-argon `argon/prompts/generate_world_repair.md @ ab1aff085977` 6,893 prompt / 6,336 output tokens	October 25, 2026 at 12:55 PM PDT	`840f468586fcc095`
world_v2 v2 ← world_v1	live	gemini-4-argon `argon/prompts/critique.md @ fe363a569e50` 869,939 prompt / 11,047 output tokens	October 25, 2026 at 1:17 PM PDT	`526ea902d59339fa`

Version	Mode	Model / prompt	Created	Output SHA-256
world_v3 v3	live	gemini-4-argon `cases/demo-full/worlds/onepass/prompt.txt @ b38120882a65` 839,188 prompt / 16,798 output tokens	October 25, 2026 at 1:56 PM PDT	`68ca90715f8b56f8`
world_v4 v4 ← world_v3	live	gemini-4-argon `argon/prompts/critique.md @ fe363a569e50` 863,985 prompt / 20,292 output tokens	October 25, 2026 at 3:05 PM PDT	`549d69915f599e85`
world_v5 v5 ← world_v4	live	gemini-4-argon `argon/prompts/critique.md @ fe363a569e50` 863,707 prompt / 18,636 output tokens	October 25, 2026 at 3:31 PM PDT	`263f8db5c71f4ea5`
world_v6 v6 ← world_v5	live	gemini-4-argon `argon/prompts/critique.md @ fe363a569e50` 859,685 prompt / 18,088 output tokens	October 25, 2026 at 3:36 PM PDT	`a8e2094a5bb7b13e`

Inputs to world_v4 (SHA-256)

File	Bytes	SHA-256
`text/docket_crash_summary.txt`	2,433	`5d549fc00b7d7d4d5d8345bd94331fe6d8e15349d927877c206641a4f2f8a11d`
`text/docket_driver_assistance_factual.txt`	27,944	`48532d2d35300245bce785c8ce90d1ac7ec094d23950c5543fca91b2707b3367`
`text/docket_har1702.txt`	163,253	`cad4efb23f6bbbc9f419168770358c67735ffb1efc955c60e8f7a33cf73ef0c1`
`text/docket_highway_att1_fl_crash_report.txt`	12,013	`054260bd9fd6cfc88835ec5ba00e3b5475de87cd41250cfa8e44eeb3af7c908c`
`text/docket_highway_att2_traffic_inventory.txt`	1,533	`57893b2b89941b1a17a9b91b4c2ff1250a51ba7cf80d06d35d0ea855980268f4`
`text/docket_highway_att5_spot_speed.txt`	2,723	`8618cef8ff1c43606450daa64ebeb60ca487a2eba609d185b3b69c827258ffae`
`text/docket_highway_factual.txt`	21,684	`576b22cb985d1ca61d380e141c064023049f28d5b4efcfca77ae0e3c559e51b6`

File	Bytes	SHA-256
text/docket_hp_att11_fhp_motor_carrier_inspection.txt	5,224	b5d7c0a86acf74deff336cf0631edf1be93300c4cdddb96a7e4371952c80a82b
text/docket_hp_att4_witness_interview.txt	37,916	6c550e1c5b44db6329b8d465349a9db865ab9d0ad8a7f521c937a39598e7663e
text/docket_hp_supplemental_errata.txt	1,467	ff948b24931f1b39204bf9404b0c4b06d8df07d2f34f4a543d69a063f8233dcb
text/docket_motor_carrier_factual.txt	18,623	a1b5c1c071eb3c16ee960799ed628fd2d83bb50363924872fb8d3084afc17f70
text/docket_preliminary_report.txt	4,171	40374499edd94e7699e737283ef16b736d9f5aae44d655780c0ce8cad36d4962
text/docket_tech_recon_factual.txt	38,007	3a942b310a84777675f8d6c3b21da5aec76e500a159fe539c2a80ce889afb024
text/docket_vehicle_att3_fhp_inspection.txt	10,027	6bba69740d264b0dfc31208a914717680ad3091b6c6d62b0bd23cb5055dbfa55
text/docket_vehicle_dynamics_study.txt	25,677	5e46779193c36b40b6326c58890cf2b00e48d032345d84e550b70ebf9df544a0
text/docket_vehicle_factual.txt	22,931	02a00c7379722725fa27f50dc837a7057fd78fdf69458745a3dbe183c1fb2d82
text/docket_vehicle_photographs.txt	2,871	865b93ad97bdd7bc1f28a953bdfebf2b41426549efa55f5ceecb6d935d4eab0d
evidence/facts.md	6,883	16a2ac328322655c10775315a2666ebef5eb8a76ddab222b6e7e8c98fcf7cfad
evidence/police_report.md	6,546	e0abbf6814d1206cc2246b864465e56c89bf71e001e5ef95b462d7bf866a2a8b
evidence/witness_A.md	1,729	57b80e791b7642498bbfb3756d46ecf878beeb5cee317c2e04313c0fc04e5c8c
evidence/witness_C.md	1,507	14873e9544ba08b9b7e61a3b3fb368ca313b76ec2ff98bc434cd73ee8cd8656e
evidence/deposition_witness_B.mp4	19,910,361	5c9dc19f396dd9760532a422edd7cfa98b2406fc9cf690f2d0625b23b3241b29
evidence/deposition_witness_D.mp4	7,313,799	9f26ce5d5e88ad0a08c1c38e40e5e0b5ef38f023959e320625e15ec25d196f6b

File	Bytes	SHA-256
evidence/exhibit4_intersection_map.png	54,576	e44f99efb5dd48300e638c39b92ad9a70c2642c31da7f66dd30b9248bf251aaa

Reconstruction file: /cases/demo-full/worlds/world_v4.js · SHA-256 549d69915f599e85a34e31244a270452db7add10971acde1c8a941abb83bf2e9. Findings and checks last updated October 25, 2026 at 3:07 PM PDT.

Generated by Sightline from case demo-full on October 25, 2026. Analysis draft for expert review — not evidence.

Reviewed by (reconstruction expert)

Date